Sanaul Haque

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Forecasting Solar Ramps from Unlabeled Sky Video: Occlusion-Aware Masked Autoencoders for the Power Grid's Hardest Forecast Horizon

Student:
Aiden Yun

Institution:
The Nueva School

Country:
United States

Email:
aiden.t.yun@gmail.com

Category:
Junior

Class:
Grade 11

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Forecasting Solar Ramps from Unlabeled Sky Video: Occlusion-Aware Masked Autoencoders for the Power Grid's Hardest Forecast Horizon

Aiden Yun

1 Introduction

Solar power is now the largest source of new electrical-generating capacity in the world. Global solar additions exceeded 600 GW in 2025, and a cumulative capacity of approximately 2.8 TW makes solar the largest generation technology of any kind [1]. But solar power has an operational weakness: when a cloud crosses over a solar plant, its power output can drop 50% to 60% in a minute [2]. Electricity supply and demand must match continuously to keep power grid frequency and voltage in safe ranges. When solar power output drops suddenly, something else has to fill the gap immediately. Without good forecasts, power grid operators must hold large reserves (spinning reserves, fast gas plants, batteries) to respond to rapid solar ramp events [3]. Accurate short-term solar power forecasting is potentially a highly cost-effective strategy to reduce both reserve requirements and outage events by giving operators advance notice to procure cheaper power sources [3].

The specific time scale most difficult for traditional numerical modeling solutions to predict is the 0–30 minute window. Numerical Weather Prediction (NWP) models use a grid spacing on the order of kilometers at a time resolution of hours; geosynchronous satellite imagery arrives every five to fifteen minutes at kilometer-scale pixel resolution [3]. Satellite imagery can predict the general area a cloud will shade, but the moment it crosses the sun typically falls between updates [3]. A ground-based sky imager is the only sensor that observes that cloud directly, providing information on its shape, motion, and position relative to the sun at sub-minute resolution [4, 3].

2 Previous Work and Model Motivation

The difficulty of forecasting at this time horizon is easy to underestimate, because the trivial forecast is right most of the time: on the sunny majority of examples, the output fifteen minutes out is nearly identical to the output now. Therefore, the field measures the success of forecasts by performance relative to smart persistence rather than by total error [3].

Measured in this way, the successes have been modest. SUNSET, a convolutional model that serves as a baseline for the SKIPP'D benchmark [5], improved upon smart persistence by 15.7% under all conditions and by 16.3% under cloudy skies [6]. More recently, SkyGPT [7] took a two-stage approach: an autoregressive model generates probabilistic future sky images, and a U-Net nowcasts power from the generated 15-minute-ahead image. While an autoregressive, purely image/video-based strategy eliminates the possibility of defaulting to persistence, errors and noise accumulate with each timestep. The method improved the continuous ranked probability score (a probabilistic metric) by 23% over smart persistence and 13% over SUNSET [7]. Like other methods that use a sky-image CNN to predict power [8], both SUNSET and SkyGPT were trained from scratch on a few years of labeled data from a single location at 64×64 resolution.

The performance bottleneck of these methods is labeled data. Supervised training requires a skyward-facing camera located next to a photovoltaic (PV) recorder or irradiance sensor, but only a handful of sites simultaneously record both [5]. Unlabeled sky video, on the other hand, is abundant: observatories and radiation-monitoring networks have captured all-sky imagery continuously for many years, but much of this imagery remains unused.

This disparity in availability of labeled versus unlabeled data motivates the question: Can we utilize unlabeled sky video to improve the accuracy of 15-minute-ahead PV forecasting? Ideally, a pretrained encoder would generalize to several different sites with less labeled data. We propose a novel architecture, SkyMAE (Sky Masked Autoencoder), to address this question. SkyMAE opts for an end-to-end approach similar to SUNSET, but will use Vision Transformers (ViTs). Previous architectures such as SUNSET used Convolutional Neural Networks, which tend to outperform ViTs on smaller datasets. However, we hypothesize that by exploiting pretraining, a ViT-based architecture can surpass a CNN-based architecture. SkyMAE will fuse PV power/irradiance with the video early, so that the Transformer can attend to both PV power and sky video simultaneously and understand their relationship. We hypothesize that by pretraining SkyMAE on unlabeled sky video data using the VideoMAEv2 architecture [9], we can improve both model performance and the ability to generalize to various solar sites with limited PV-labeled sky imagery data.

Our research proposal is summarized below:

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1. We propose SkyMAE, an end-to-end Vision-Transformer-based architecture that can learn from unlabeled sky video, overcoming the primary limiting factor for Vision Transformers—insufficient labeled data.

2. We propose occlusion-aware masking: by excluding tubelets containing non-sky pixels from both the encoder input and the reconstruction loss, the method enables training on large, old datasets with sun-blocking artifacts.

3. We propose embedding power history as tokens, which are temporally aligned to video tubelet tokens, to enable the encoder to, for example, determine which specific cloud was responsible for a given drop in power by using attention to relate the two features.

4. We alter the solar forecasting paradigm from predicting absolute solar values to predicting the change between current and future (the residual), thereby greatly reducing any tendencies of the model to default to persistence.

3 Methods

3.1 Pretraining

Models pretrained using self-supervised learning can learn the context of data and generate feature-rich representations to improve performance over a purely supervised model. For videos, VideoMAE [10] and, more recently, VideoMAEv2 [9] are considered state-of-the-art. VideoMAE uses tube masking, splitting 224×224 frames into 16×16 patches and randomly selecting a high proportion of these patches to be masked across all 16 frames. A Vision Transformer (ViT) encoder generates feature-rich embeddings for the non-masked patches,

which are passed into a ViT decoder, whose goal is to reconstruct the original video. The model is optimized end-to-end to minimize pixel-wise mean squared error. Once training is complete, the decoder is discarded, leaving an encoder that understands the data and can generate useful embeddings.

For pretraining, we plan to use VideoMAEv2 base-patch16-224 with embedding dimension $D = 768$ and a 90% masking ratio.

3.2 Fine-tuning

3.2.1 Data

The goal of model training during fine-tuning is to forecast a single value: the photovoltaic output 15 minutes in the future.

We plan to incorporate multimodal input: 16 minute-separated frames of past sky video paired with the corresponding 16 irradiance/PV $(P_{t-15}, P_{t-14}, \ldots, P_t)$ values at the time of each frame.

Before being input into the model, the P values are normalized utilizing the standard deviation and mean of the P values in the training set:

$$\tilde{P}_j = \frac{P_j - \mu_P}{\sigma_P}$$

We theorize that having the model predict the P value directly can cause it to default to persistence (predicting that the future P does not change from the current P). Since the model is provided with P_t, and for sunny days, which comprise the majority of samples, $P_t \approx P_{t+15}$, the model can simply return P_t and score a low loss for most samples without improving on the intended task: abrupt change in power output caused by turbulent weather and cloud cover (ramp events). We will attempt to combat this by having the model predict the residual relative to persistence, or in other words, $P_{t+15} - P_t$, which discourages the model from over-attending to P_t.

We plan to normalize this target to improve training stability:

$$\delta = \frac{P_{t+15} - P_t - \mu_r}{\sigma_r},$$

where σ_r and μ_r are the standard deviation and mean of the training set residual, respectively.

3.2.2 Model Architecture

15 We propose the model architecture shown in Figure 1. The model utilizes cube embedding [11, 10] for the sky video, using a Conv3D layer to split the $(B, 16, 224, 224, 3)$ input into 1568 tubelets of shape $(2, 16, 16, 3)$ and project each of these to a D-dimensional vector, creating a matrix of shape $(B, 1568, D)$.

The PV values are split into 8 consecutive pairs to temporally match the sky video tubelets, which each span 2 minutes. This creates a matrix of shape $(B, 8, 2)$, which is passed through an MLP, projecting it to shape $(B, 8, D)$ such that it can be concatenated with the sky video tubelet embeddings.

Prior to concatenation, learned positional embeddings are added to both the sky video tubelet and PV pair embeddings. The two embedding matrices are then concatenated along the token dimension and passed through a Transformer encoder, whose weights are initialized from pretraining. The encoder produces feature-rich embeddings that are then passed to a regression head, replacing the typical ViT/ViViT [11] classification head and its cross-entropy loss with MSE loss.

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The loss is calculated as the mean squared error between the model's prediction, $\hat{\delta}$, and the ground truth

$$\delta = \frac{P_{t+15} - P_t - \mu_r}{\sigma_r}.$$

At inference time, the PV prediction is de-normalized:

$$\hat{P}_{t+15} = P_t + \mu_r + \sigma_r \hat{\delta}.$$

Although sun blockers obscure the sun and its immediate surroundings, the data still has utility potential. We propose identifying blocker pixels with a low-intensity threshold, since they are black against clouds, the sky, and the sun, and excluding every tubelet that contains them from both the encoder input and the reconstruction loss, so the model never needs to reconstruct the blocker. Because the blocker tracks the sun, the excluded region moves from sample to sample, leaving no fixed blind spot.

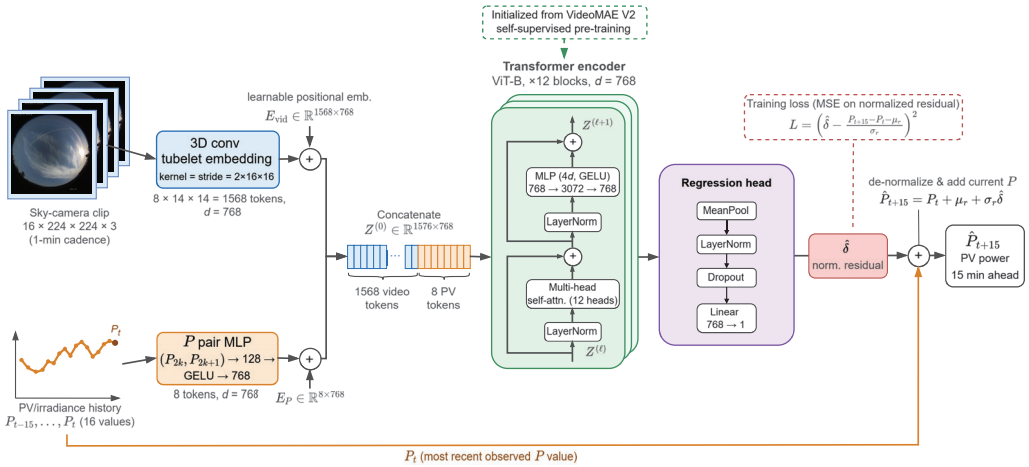


Figure 1: Fine-tuning architecture

4 Experiments

4.1 Training

We plan to train using data from the University of Hertfordshire Bayfordbury Observatory, which records sky video at a cadence of 1 frame per 19 seconds [12]. The camera has a circular sun-blocker attachment that is present in about half of the recorded days in the dataset. Since the time between frames is not exactly 20 seconds, sampling every 3 frames for a total of 16 frames would result in a roughly 45-second deficit from the intended 15-minute interval. We instead use cadence-matched sampling, selecting the closest frame to

each of the minute-separated timestamps starting from the oldest frame up to the current frame (e.g., 10:01:15, 10:02:15, 10:03:15, . . .). This means that, in the worst case, two frames can be spaced 50 seconds or 70 seconds apart. We hypothesize that this is the better strategy, as it preserves the forecast horizon.

4.2 Evaluation and Expected Outcomes

We will test and evaluate on SKIPP'D [5], which records PV power in kW, and SolarVision Almería [13], which records global horizontal irradiance in W/m^2. Evaluating on both datasets tests whether the approach transfers across sites, cameras, and target variables. For SKIPP'D we will use the test samples selected by Nie et al. so that results are directly comparable to published baselines. For Almería we will hold out complete months spanning all four seasons. All hyperparameters, thresholds, and splits will be fixed on the validation set before any test-set evaluation, and every configuration will be trained with three random seeds, with the mean and standard deviation reported.

We will report MAE, RMSE, and forecast skill relative to smart persistence, FS $= 1 -$ RMSE$_{\text{model}}$/RMSE$_{\text{SP}}$, the field's standard yardstick [3, 6]. Smart persistence holds the clear-sky index fixed over the horizon, $\hat{P}_{t+15} = k_{\text{clr}}(t)\, P_{\text{clr}}(t + 15)$, where $P_{\text{clr}}(t)$ is the output expected under a cloudless sky and $k_{\text{clr}}(t) = P_t / P_{\text{clr}}(t)$. Our benchmarks will be the state-of-the-art supervised methods SUNSET [6] and SkyGPT [7].

We expect sky-video pretraining to outperform both no pretraining and generic video pretraining, with the largest gains on the ramp and cloudy subsets; the fused model to beat either modality alone; SkyMAE to exceed SUNSET in MAE, with RMSE and the ramp subset as the harder test, since RMSE is dominated by the rarest, largest ramps; occlusion-aware masking to recover the value that discarding sun-blocker days forfeits; and the pretrained model's advantage to widen as labels shrink. Negative results are useful too: if standard masking matches occlusion-aware masking, the blocker is a smaller obstacle than assumed. If MAE improves while RMSE does not, then the model has stopped mimicking smart persistence but still misses the large ramp events.

5 Significance

Every sky-image forecaster to date has been trained at the few sites where a camera and a power sensor happen to coincide, while monitoring networks and observatories have recorded all-sky video for years with no co-located sensor and a sun-blocker in every frame. If our hypotheses hold, an encoder pretrained on footage the field has been discarding will need only a small labeled set at a new site, making it practical to implement camera-based forecasting for solar farms. The stakes scale with the terawatts of solar now being installed [1]: better 15-minute forecasts reduce reserve requirements and outage events. Our new methods also generalize beyond solar. Occlusion-aware masking applies to any imaging archive with a known moving obstruction, and cadence-matched sampling applies to any archival video whose frame rate does not match the model's input.

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